

ORBIT NORTH

Canadian Summer Space Intensive

GNG 3100 · SYS 5186

Summer 2026

Design a CubeSat mission. Command a live mission-control simulation. Build a career in space — in a 12-day intensive program.

6 – 21 August 2026 · University of Ottawa · uOttawa Cyberrange Mission Control Facility

At a glance

12-day intensive program	6 – 21 August 2026, weekdays, 09:00 – 17:00, in person
Mission · end-to-end	from Canadian context to operator console
2 codes · UG and Grad	GNG 3100 (UG) · SYS 5186 (Grad)
Prior space experience	not needed

ORBIT NORTH takes you from the Canadian space ecosystem and its regulatory environment, through a full preliminary CubeSat design in a Concurrent Design Facility, to the seat of a mission-control console — in twelve in-person days at the University of Ottawa. Built for students with the drive to do something real.

What you'll do

WEEK 1 — CANADIAN CONTEXT, CDF INTRO & CONCEPT DEVELOPMENT · 6 – 7 AUGUST 2026

Two days that ground you in the Canadian space ecosystem and put you straight to work. Thursday morning covers the past, present, and emerging Canadian sector; Thursday afternoon brings a short regulatory teaser and your first session in the SpaceCDF tool. Friday is mission architecture and concept work in the CDF, with the team's risk register opened at end of day.

Canadian sector · SpaceCDF · Mission concept · Risk register

WEEK 2 — MCR, REGULATORY & CDF INTENSIVE · 10 – 14 AUGUST 2026

Monday morning is the **Mission Concept Review**; Monday afternoon covers the Canadian regulatory landscape — RSSSA, ITU/ISED spectrum coordination, export controls, debris mitigation, registration. The remainder of the week is the SpaceCDF intensive: requirements baseline, power, AOCS and thermal, communications, structure and propulsion, integration, V&V, cost, and risk closure — converging on a **Preliminary Design Review** on Friday afternoon.

MCR · RSSSA · ISED · Subsystem design · Trade studies · PDR

WEEK 3 — OPERATIONS & MISSION SIMULATION · 17 – 21 AUGUST 2026

Two days of spacecraft operations training at the uOttawa Cyberrange — ground systems, flight dynamics, mission-control system, planning tool, procedures. Then two days running a live simulation: take the operator seat while another team observes and documents, then swap. Work nominal operations and inject contingency in real time. Closes with a structured debrief and final mission review.

Ground systems · Flight dynamics · Anomaly response · Mission debrief

What you'll be able to do

- Situate a mission concept in the Canadian space ecosystem.
- Navigate Canadian satellite licensing, export controls, and RSSSA obligations.
- Take a discipline role in a Concurrent Design Facility and deliver a complete preliminary CubeSat mission across all principal subsystems.
- Maintain a risk register and a planning baseline through the design week.
- Execute spacecraft operations procedures in a mission-control environment.
- Respond to anomalies and contingency scenarios in a live simulation.

Two codes · differentiated outcomes

Code	Level	Notes
GNG 3100 — Topics in Engineering I	Under-graduate	No prior space background required.
SYS 5186 — Advanced Topics in Systems Engineering	Graduate	Additional outcomes around quantitative trade-off evaluation, sub-area leadership, and literature-grounded synthesis.

Who's involved

University of Ottawa Faculty of Engineering · School of Engineering Design and Teaching Innovation (SEDTI) · uOttawa Cyberrange.

Interested? Get in touch.

Dr Franz Newland · SEDTI, University of Ottawa fnewland@uottawa.ca

→ Express your interest / Apply via Brightspace.

ORBIT NORTH · GNG 3100 / SYS 5186 · Summer 2026